

BINLEVEL · DESIGN PRECIS

BinLevel design precis

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DOCUMENT CONTROL

Document ID	BNL-PRC-001	Version	0.7
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/binlevel		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Open decisions of 2026-10-02 carried in: steel containers out of the first pilot, galvanic isolation, data layer, fill rate left to the server, first partner to approach
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Design for construction (BNL-DDR-003); 55 mm box, box fixings and printed sensor mounts added; mass, envelope and cost from BNL-CAL-001 v0.3
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Design choices adopted for TRL 3 (BNL-DDR-001); numbers checked against BNL-CAL-001; 11-byte payload, nanopower regulator, 5 min temperature reads and cell strap added
0.2	2026-09-25	Amish Chadha	

Concept for TRL 2 with components,
first-order numbers, safety and open
questions

0.1	2026-09-25	Amish Chadha	Initial scaffold
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BinLevel design precis

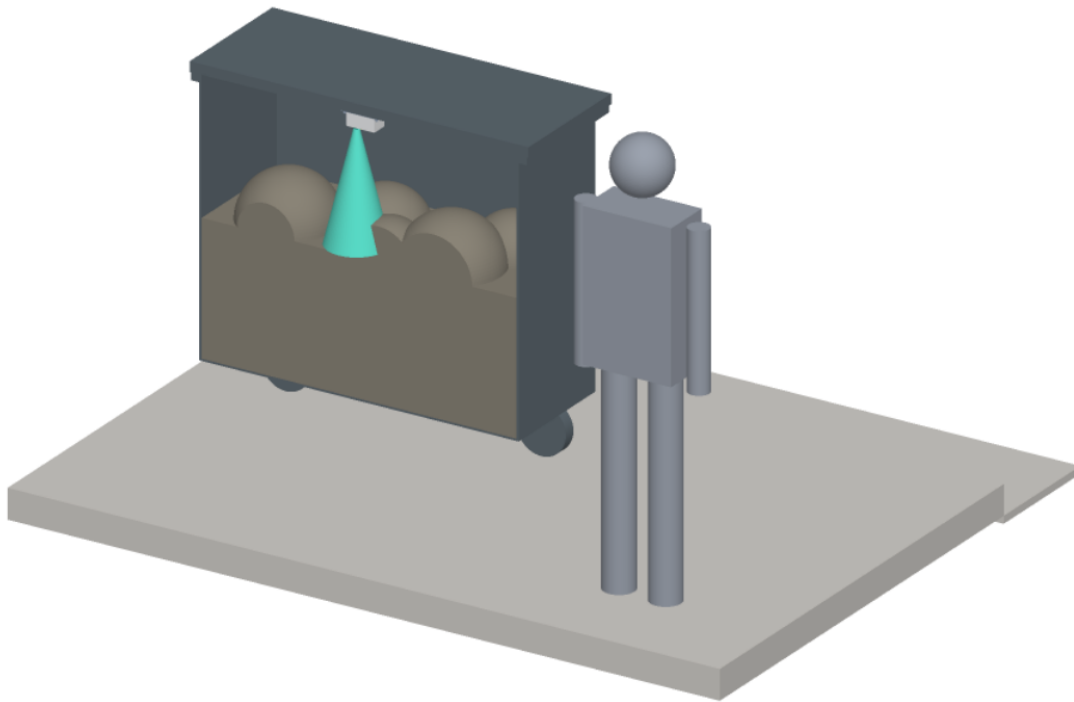
■ Summary

BinLevel is a small sealed box bolted under the lid of a communal or public bin. Every 15 minutes it measures the distance down to the waste with a sealed ultrasonic transducer, backed by a time-of-flight (ToF) sensor for the top of the bin, and turns it into a fill percentage. It sends the level, temperature, tilt events and battery state by LoRaWAN every hour (every 2 hours at SF12), or at once when the bin passes a set level. A primary lithium C cell would last about 16 years in the worst radio case on paper, so seals and plastics, not the cell, set the 10-year design life. Parts cost is \$59.00 (indicative), within the \$60 value-engineering target, and the unit weighs about 350 g, inside the 400 g target (BNL-CAL-001). The design is constructable (BNL-DDR-003), and the prototype build plan BNL-BLD-001 shows how to build it. It carries no camera and no microphone.

BinLevel

github.com/BoujeeEnjinia1701/binlevel

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: 1100 L container, shown in section, waste at about 55 % fill, pavement, 1.75 m person, teal cone: ultrasonic beam, illustrative for scale

■ How it works

1. **Measure.** The microcontroller wakes every 15 minutes, switches on the ultrasonic transducer and takes five pings, then reads the ToF sensor. It keeps the median ultrasonic distance and uses the ToF reading when the surface is closer than the ultrasonic blind zone (about 0.25 m, estimate).

2. **Interpret.** Distance is converted to fill percentage against the empty depth stored at installation. A fill above the threshold (default 80 %) or an abnormal temperature, read every 5 min, triggers an immediate uplink. The heat rule is 70 °C, or a rise of 15 K in 15 min counted only while the unit is above 50 °C, so a sunny lid does not raise false alarms.
3. **Sense events.** A low-power accelerometer wakes the controller when the bin is tipped (emptying confirmed) or when the lid is opened (usage count), so the service gets proof of collection without a driver app.
4. **Report.** A LoRaWAN Class A uplink of 11 bytes carries fill %, raw distance, temperature, event counters and battery voltage. Any LoRaWAN network server can receive it: the lab's TwinKit gateway, The Things Network or a city network.
5. **Plan.** The server passes levels to a route planner, which schedules the bins that are full or will be full before the next round (Figure 2).

BinLevel: data flow (estimates; fill level only, no images or audio)

github.com/BoujeeEnjinia1781/binlevel

CONCEPT, NOT FOR FABRICATION



■ Main components

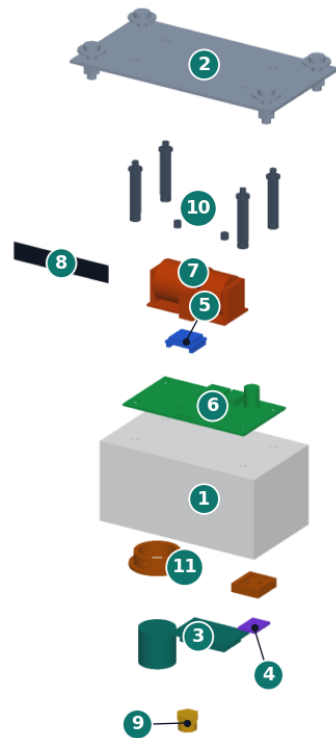
Numbers match the exploded view (Figure 3) and `bom/bom.csv`. Main dimensions and interfaces are on the general arrangement drawing BNL-DWG-001 (`cad/drawings/BNL-DWG-001.pdf`), generated from `cad/src/model.py`.

No.	Component	Role
1	Enclosure, IP67 ABS or PC, about 115 x 65 x 55 mm, cover facing down	Protects electronics; holes for transducer, ToF window and vent in the cover
2	Aluminium bracket plate, 2 mm 5052 class, and four stainless tamper-resistant M6 x 20 bolts on a 130 x 60 mm pattern	Fixes the unit under the lid; bolts pass through the lid
3	Sealed 40 kHz ultrasonic transducer (JSN-SR04T class) with driver	Main range sensor, about 0.25 to 1.5 m (estimate)
4	Near-range ToF sensor (VL53L1X class) behind a window	Top of bin and overfill, about 0.03 to 0.4 m (estimate)
5	LoRaWAN module, RAK3172 (STM32WL)	Controller and radio in one module, as in FieldNode
6	Carrier board with accelerometer, temperature sensor, nanopower regulator, load switch and buffer capacitor (a prototyping board with breakouts for the first prototype)	Joins the parts; powers sensors only while measuring; keeps the module below 3.6 V
7	Li-SOCl ₂ primary cell, C size, 3.6 V, about 8.5 Ah nominal, with strapped holder and fuse	Multi-year energy store
8	Flexible 868/915 MHz antenna	Radio antenna inside the enclosure (external lid antenna for steel bins is a later option; not in the first pilot)
9	Gasket and vent membrane	Keeps water and condensation out
10	Box fixings: four press-in studs in the plate, sealing washers and standoffs	Hold the box to the plate and carry the board
11	Printed probe collar and ToF holder, window disc	Hold the sensors in the cover

BinLevel: exploded viewgithub.com/BoujeeEnjinia1701/binlevel

CONCEPT, NOT FOR FABRICATION

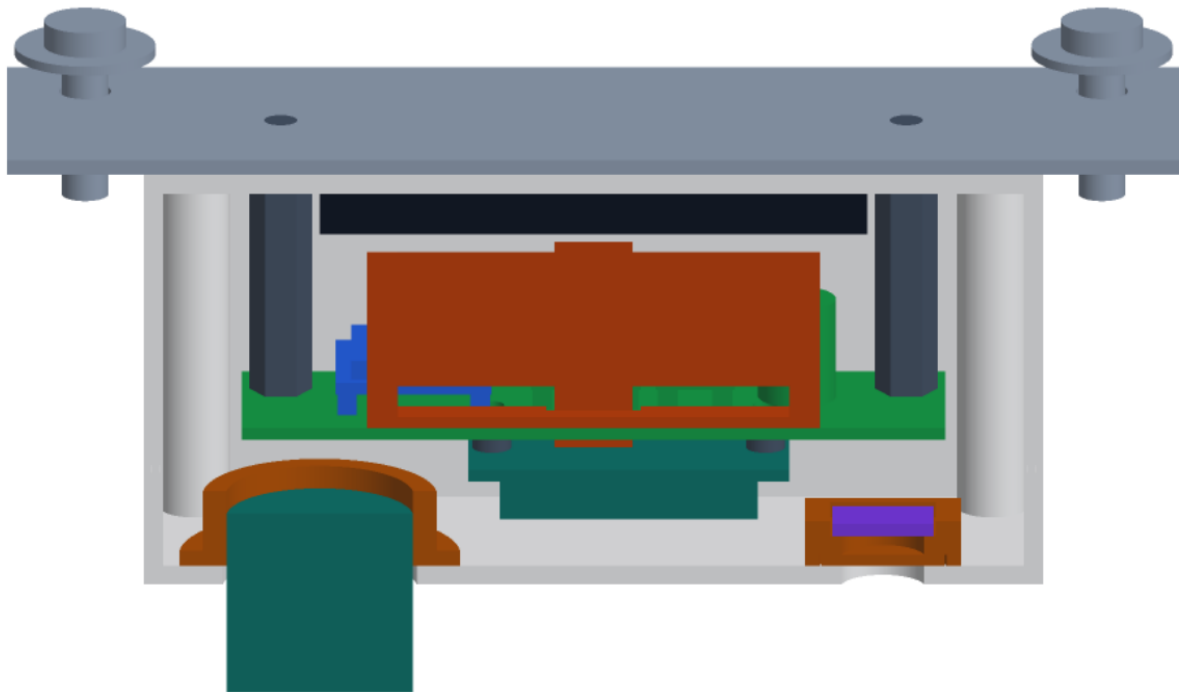
- 1 Enclosure, IP67 ABS or PC
- 2 Aluminium bracket plate and tamper bolts
- 3 Ultrasonic transducer and driver
- 4 Near-range ToF sensor
- 5 LoRaWAN module on breakout
- 6 Electronics board and small parts
- 7 Li-SOCl₂ C cell, holder and strap
- 8 Flexible antenna
- 9 Gaskets and vent
- 10 Box fixings: studs, standoffs, seals, screws
- 11 Printed sensor mounts and window



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

BinLevel: cutawaygithub.com/BoujeeEnjinia1701/binlevel

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

■ First-order numbers

The numbers below are from the sizing note BNL-CAL-001, which checks every requirement against the parametric model `cad/src/model.py`. They are paper estimates, not measurements.

Energy budget

Sleep current is $6.0\ \mu\text{A}$ for the whole unit (module $2.0\ \mu\text{A}$, accelerometer $2.0\ \mu\text{A}$, temperature sensor $0.5\ \mu\text{A}$, regulator and leakage $0.5\ \mu\text{A}$, buffer capacitor $1.0\ \mu\text{A}$). Each ranging cycle takes $7.5\ \text{mAs}$; each uplink transmits at $+14\ \text{dBm}$ at an assumed $45\ \text{mA}$. Usable capacity is taken as 60 % of nominal, with self-discharge of 1 % of nominal a year.

Table 1. Daily charge use and life, with two alert uplinks a day (BNL-CAL-001, section C).

Item	SF9, hourly uplink	SF12, uplink every 2 h
Sleep (6.0 μ A x 24 h)	0.144 mAh	0.144 mAh
Ranging (96 cycles) and temperature reads	0.201 mAh	0.201 mAh
Uplinks	0.086 mAh (26 x 0.21 s airtime)	0.278 mAh (14 x 1.48 s airtime)
Self-discharge (C cell)	0.233 mAh	0.233 mAh
Total	0.664 mAh per day	0.856 mAh per day
Life on C cell (5.1 Ah usable)	21.0 years	16.3 years
Life on AA cell (1.56 Ah usable)	8.5 years	6.2 years

The cell is not the limit on life; seals, the transducer membrane and the enclosure plastic under UV and heat are. The design target is therefore stated as 10 years (R4). The AA cell would meet the 5-year minimum but not the 10-year target, which is why the C cell is adopted (DDR-001, D3).

A fresh Li-SOCl₂ cell rests at 3.67 V, above the RAK3172 module's 3.6 V maximum ([RAK3172 datasheet](#)), so the carrier includes a nanopower 3.3 V regulator. The cell carries the transmit pulse; a 1,000 μ F low-leakage capacitor covers the voltage delay of a passivated cell.

Radio airtime

The payload is 11 bytes (it was 12 at TRL 2), so that it fits the 11-byte limit of the slowest US915 data rate. It takes about 0.21 s at SF9 and about 1.48 s at SF12 (125 kHz). Hourly uplinks with two alerts use about 5.4 s per day at SF9, well inside TTN's 30 s daily fair use, but 38.6 s at SF12, which would exceed it. SF11 hourly uses 21.4 s. The firmware therefore stretches the routine interval to 2 hours at SF12 only (20.8 s per day) and keeps threshold alerts immediate (DDR-001, D6, as refined by DDR-002).

Link

With a 10 dB fade margin and an urban path-loss model, a sensor in an HDPE container reaches a gateway about 0.8 km away at SF9 and 1.3 km at SF12. Inside a steel container the range falls to about 0.35 km even at SF12; an external lid antenna restores 1.3 to 2.1 km. Steel containers are left out of the first pilot, which uses plastic containers only (decided by Amish, 2026-10-02; DDR-001, O1); the antenna is revisited once the link loss inside a steel container has been measured.

Measurement

- The transducer face is 1.00 m above the floor of an 1,100 L container. At 80 % fill the surface is 0.20 m below the face, inside the ultrasonic blind zone (about 0.25 m, estimate), which is why the ToF sensor covers the top 0.4 m. The two sensors overlap between 60 % and 75 % fill.
- The instrument error, with speed of sound compensated from the on-board temperature, is about 10 mm for the ultrasonic sensor and 20 mm for the ToF sensor, against the ± 101 mm (± 10 % of depth) accuracy target. Accuracy on real waste is limited by the surface, not by the sensor.
- An ultrasonic beam of 15° half-angle (assumed) covers a patch about 0.54 m across at the floor. Mounted at the lid center, the beam clears the walls for half-angles up to about 26°.
- The temperature is read every 5 min (not only with each 15 min ranging), so a heat alert reaches the server within about 7.5 min.

Mass and cost

- Mass about 350 g: enclosure 94 g, 2 mm aluminium plate 63 g, bolts, nuts and washers 36 g, box fixings 22 g, cell, holder and strap 59 g, transducer, board, module, printed mounts and small parts about 76 g. This is inside the 400 g limit of R16. The 1.5 mm stainless bracket of BNL-PRC-001 v0.3 weighed 183 g and made the unit about 435 g; Amish chose the aluminium bracket on 2026-09-25 (DDR-002).
- Envelope below the lid 150 x 80 x 71.5 mm.
- Parts cost \$59.00 at prototype quantities (indicative prices; see `bom/bom.csv`), inside the \$60 value-engineering target. The gateway is shared across many bins and not included.

■ Key design choices (decided by Amish, 2026-09-25)

Each choice below was decided by Amish on 2026-09-25 (go with recommendation) and is recorded in [decisions/0001-trl2-review-decisions.md](#) (BNL-DDR-001) and [decisions/0002-recommendations-accepted.md](#) (BNL-DDR-002).

1. **Ultrasonic plus ToF, rather than ultrasonic alone or ToF alone (D2).** Ultrasonic tolerates dust and dirt on the face and is the method used by commercial bin sensors ([Sensoneo](#)). Low-cost sealed transducers cannot see the top 0.25 m, where the full threshold lies, so a ToF sensor covers that zone for about \$5.
2. **Lid underside mounting (D8).** Keeps the unit out of the waste, out of sight and away from vandals. The cost is that lid opening moves the sensor, which the accelerometer detects and skips.
3. **Primary Li-SOCl₂ C cell, not rechargeable or solar (D3).** No charging electronics, no panel to break on a tipped bin, and very low self-discharge.
4. **LoRaWAN on an STM32WL-class module (D4).** The RAK3172 shares the radio core and firmware approach adopted for FieldNode, so decoders and gateway set-up are common. The RAK3172-T variant (-40 to 85 °C) is named for sites colder than -20 °C (DDR-002). NB-IoT would avoid gateways but needs a SIM subscription per bin.
5. **Stock enclosure at TRL 2 to 3 (D9).** Keeps the build garage-friendly; a potted or IP69K design is a later option.
6. **Relationship to FieldNode and TwinKit (D5).** BinLevel does not use the FieldNode enclosure, panel or LiFePO₄ cell: at \$126 the FieldNode core costs more than twice the BinLevel value-engineering target and would not fit under a lid. It reuses FieldNode's radio module family, payload conventions and decoder so both sit on the same TwinKit gateway.
7. **First target container (D1), thresholds (D6) and temperature alert (D7).** The 660 to 1,100 L communal container is the design case; the default fill threshold is 80 %, the routine interval 1 h (2 h at SF12 only, DDR-002), and the heat alert 70 °C or a 15 K rise in 15 min. BNL-CAL-001 found that the ungated rate-of-rise trigger could fire when sun breaks through cloud, so under DDR-002 the rise counts only above 50 °C; the requirement R6 upper limit is also raised to 70 °C to match hot dark lids.
8. **Aluminium bracket (DDR-002).** A 2 mm 5052-class aluminium plate replaces 1.5 mm stainless, taking the unit from about 435 g to 334 g so that it meets R16 (350 g after the design for construction, BNL-DDR-003). The bolts stay stainless; galvanic isolation between them and the plate: nothing is fitted for the prototype, and for deployments the plate is anodised before the studs are pressed in (decided by Amish, 2026-10-02).

■ Safety

Safety: The Li-SOCl₂ cell is a primary lithium cell with high energy density. Never charge it, short it, crush it or heat it above its rated limit; a damaged cell can vent toxic, corrosive gas or catch fire. Fit a fuse or PTC in series, use a holder that prevents reverse insertion, and dispose of spent cells through a battery recycler, never in the bin itself.

Safety: Bins can hold sharp objects, needles, broken glass and biological waste. Fit and service sensors only on emptied and cleaned containers, wear cut-resistant gloves and eye protection, and keep drill swarf out of the container. Drilling a lid leaves sharp edges; deburr them.

Safety: Bins can catch fire from hot ash or discarded batteries. The temperature alert in R9 is a maintenance aid and must not be relied on as fire detection. The unit itself contains a lithium cell, which adds fuel to a bin fire.

Safety: Keep clear of the lifting mechanism of collection trucks when installing or testing on containers in service.

■ Open questions

- [] Can a sealed low-cost transducer survive years of condensation and food acids, or is a potted industrial transducer needed?
- [] Does the ToF window fog or foul too quickly to be useful? A heater is ruled out (a 20 mW heater would use 156 times the energy budget, BNL-CAL-001); is a hydrophobic coating or a wiper enough?
- [] What is the real link loss inside HDPE and steel containers? BNL-CAL-001 assumes 10 dB and 30 dB. Steel containers are left out of the first pilot until this is measured (decided 2026-10-02; BNL-DDR-001, O1).
- [] Is IP67 enough for the first partner's washing practice? The first partner to approach is a municipal waste service or its contractor using EN 840 plastic communal containers in a city with public LoRaWAN coverage, by default a European city on EU868; its washing practice is asked before committing (decided 2026-10-02).
- [x] Which route planner and data layer should the pilot use? TwinKit as the data layer, feeding a standard open-source vehicle routing tool; a city system only if the pilot partner already runs one (decided 2026-10-02).
- [x] Should the payload also carry a rolling fill-rate estimate? No: the server estimates the fill rate from the history, and the spare payload byte stays free (decided 2026-10-02).